

YUCATAN BASIN BUOY 42056, Mexico

WMO: 997395

Lat: 19.820N

Lon: 84.945W

Elev: 0

StdP: 101.33

Time Zone: -6.00 (W06)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	22.6	12.9	9.3	22.4	13.9	9.9	23.1	14.5	23.3	13.3	24.1	8.9	40	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.6	30.2	26.0	30.1	26.0	29.9	26.0	27.2	29.1	27.0	29.1	26.7	29.1	4.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.2	28.8	26.3	21.8	28.8	26.1	21.5	28.8	86.0	29.1	84.9	29.1	83.9	29.1	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.7	11.6	10.7	20.9	30.6	0.9	0.7	20.3	31.1	19.7	31.5	19.2	31.9	18.5	32.4		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	25.3	25.5	25.6	26.6	27.4	28.1	28.6	28.9	28.7	28.0	26.7	26.0
	DBStd	1.59	1.27	1.11	1.11	0.82	0.67	0.74	0.63	0.68	0.78	0.92	1.04	1.26
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6254	476	435	483	498	539	544	576	587	562	557	500	497
	CDD18.3	3212	217	202	225	248	281	294	318	328	312	299	250	238
	CDH23.3	34946	1623	1575	1753	2433	3106	3610	4082	4349	4081	3644	2545	2146
	CDH26.7	9244	38	16	29	229	678	1233	1611	1876	1697	1213	418	206
Wind	WSAvg	6.6	7.5	7.1	7.1	6.9	6.3	6.6	6.0	5.4	5.3	6.6	7.4	7.2
Precipitation	PrecAvg	877	46	26	21	13	62	104	56	77	94	179	139	58
	PrecMax	1340	128	83	49	47	222	234	152	159	241	517	395	207
	PrecMin	456	12	4	6	0	4	6	9	19	20	25	26	12
	PrecStd	252	30	18	12	10	63	65	40	41	63	132	89	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.7	27.2	27.6	28.7	28.9	29.7	30.2	30.3	30.5	30.1	29.2	28.5
		MCWB	23.9	23.8	23.9	26.2	25.5	26.2	26.3	26.1	25.6	26.0	25.7	25.0
	2%	DB	27.3	27.1	27.2	28.2	28.7	29.6	29.9	30.2	30.2	29.7	28.9	28.2
		MCWB	23.8	23.7	23.8	25.6	25.3	26.2	26.2	26.1	25.8	25.9	25.6	24.9
	5%	DB	27.1	27.0	27.1	28.0	28.5	29.3	29.7	30.1	30.1	29.5	28.6	28.0
		MCWB	23.7	23.6	23.8	25.2	25.3	26.1	26.1	26.0	25.9	25.8	25.3	24.8
	10%	DB	26.9	26.7	26.8	27.7	28.3	29.2	29.5	29.9	29.8	29.3	28.3	27.7
		MCWB	23.6	23.4	23.6	24.8	25.1	26.1	26.1	26.0	25.9	25.7	25.1	24.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.1	25.4	27.2	26.7	27.6	27.2	27.2	27.6	27.1	26.7	26.0
		MCDB	26.2	26.1	26.8	28.4	28.2	28.9	29.1	29.5	29.1	29.3	28.4	27.8
	2%	WB	25.6	25.2	25.0	26.6	26.2	27.2	27.1	27.0	27.1	26.7	26.2	25.6
		MCDB	26.2	26.5	26.7	28.2	28.1	28.9	29.2	29.6	29.3	29.1	28.4	27.7
	5%	WB	24.9	24.5	24.6	25.7	26.0	27.0	26.8	26.7	26.7	26.5	26.0	25.4
		MCDB	26.3	26.6	26.7	27.8	28.1	28.8	29.2	29.5	29.3	29.1	28.4	27.7
	10%	WB	24.3	24.1	24.2	25.2	25.7	26.7	26.6	26.5	26.5	26.2	25.6	25.1
		MCDB	26.4	26.6	26.7	27.4	28.0	28.8	29.1	29.5	29.3	29.0	28.2	27.6
Mean Daily Temperature Range	5% DB	MDBR	1.3	1.2	1.1	1.0	1.2	1.4	1.4	1.6	1.7	1.8	1.6	1.4
		MCDBR	1.6	1.2	1.3	1.1	1.3	1.4	1.6	1.6	1.9	1.8	1.9	1.7
		MCWBR	1.5	1.5	1.5	1.2	1.3	1.4	1.6	1.5	1.6	1.6	1.7	1.6
	5% WB	MCDBR	1.3	1.5	1.3	1.1	1.4	1.4	1.6	1.7	1.8	1.8	2.0	1.5
		MCWBR	1.6	1.9	1.7	1.5	1.4	1.5	1.6	1.7	1.8	1.7	1.7	1.7
Clear-Sky Solar Irradiance	taub	0.379	0.384	0.384	0.428	0.447	0.475	0.489	0.447	0.420	0.408	0.392	0.387	
	taud	2.550	2.528	2.523	2.374	2.368	2.316	2.253	2.416	2.473	2.531	2.535	2.545	
	Ebn at Noon	900	914	924	881	853	822	812	852	876	879	879	879	
	Edh at Noon	96	103	107	125	125	130	139	119	111	101	97	94	
All-Sky Solar Radiation	RadAvg	4.34	5.34	6.23	6.87	6.63	5.99	6.64	6.37	5.77	4.84	4.42	4.12	
	RadStd	0.30	0.30	0.21	0.24	0.51	0.45	0.41	0.28	0.43	0.56	0.28	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page